Revision

Monday, November 10, 2025 3:13 PM

Theory of Computation – Revision Problem Set

Problem 1 – Smart Door Authentication (Applied DFA)

A smart door uses a 5-symbol code entered on a keypad.

The alphabet is `{O, P, E, N, 1, 2}`.

Rules:

- The code must **start with a letter** and **end with a digit**.
- The letters in the first four positions must form one of the following:
  - `OPEN`, `OPEP`, or `OPPP`.

Tasks

a) Construct a **DFA** that accepts all valid codes. Clearly define states and transitions.

b) Give a **regular expression** describing the same language. $OP(EN|EP|PP)\cdot(1|2)$

c) Explain why the language is **regular** even though it encodes a word constraint.

Problem 2 – NFA → DFA Conversion

Given the NFA N over {a, b}:

State	a	b	ε
→q0	{q0, q1}	-	{q2}
q1	-	{q2}	-
*q2	{q2}	{q1}	-

a) Apply the **subset construction** to obtain an equivalent DFA.

b) Draw the transition diagram of your DFA.

Problem 3 – Binary Stream Filter (Regex → NFA)

A hardware data filter accepts binary streams with the following property:

- The total number of **1s** in the string must be **even**.
- Each string must **end with "01"**.

Tasks

a) Write a **regular expression** describing this set.

$[A-Z]$

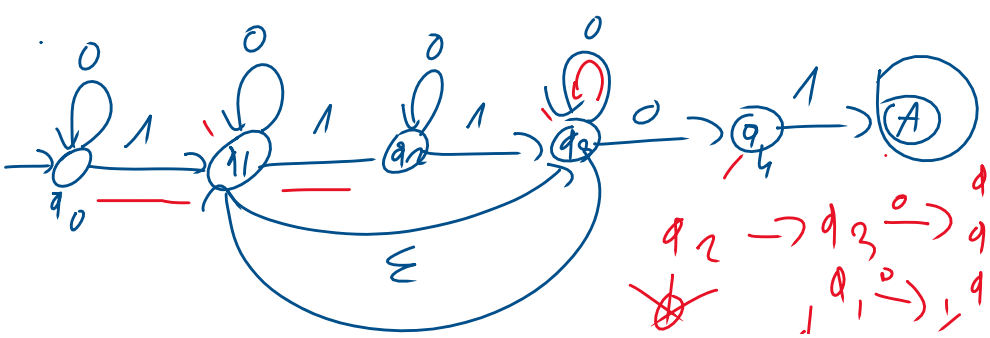
$[A-Z\,a-z\,0-9]$

$(0^*10^*10^*)^*01$

$\cancel{(0^*10^*)^*01}$

$0^*1(0^*10^*10^*)^*01$

b) Construct an **NFA** that accepts this language.



$q_0 \xrightarrow{1} q_1 \xrightarrow{0} q_1 \xrightarrow{1} q_2$

$q_3 \xrightarrow{0} q_3 \to \phi$

$q_4 \to A$

$q_2 \to q_3 \xrightarrow{0} q_4 \to A$

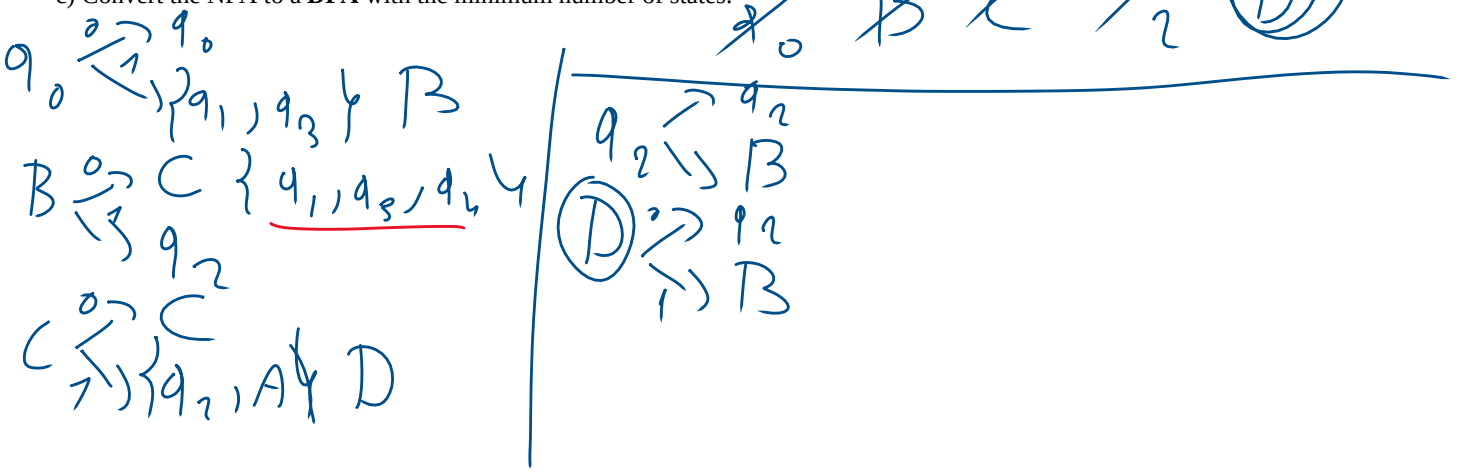
$q_3 \to X$

$q_1 \xrightarrow{0} q_1$

101101

101101 1

c) Convert the NFA to a **DFA** with the minimum number of states.



Problem 4 – Non-Regular Language (Pumping Lemma)

Prove that the following language over {0, 1} is **not regular**:

L={ $0^n1^m0^n$|n,m≥1 }
Use the **Pumping Lemma**.

Problem 5 – Regular vs Context-Free

Consider the language

L={ $a^ib^jc^k$ | i=j or j=k }
a) Show that L is **not regular**.

Let S = a^p b^p c^q j is not equal to k

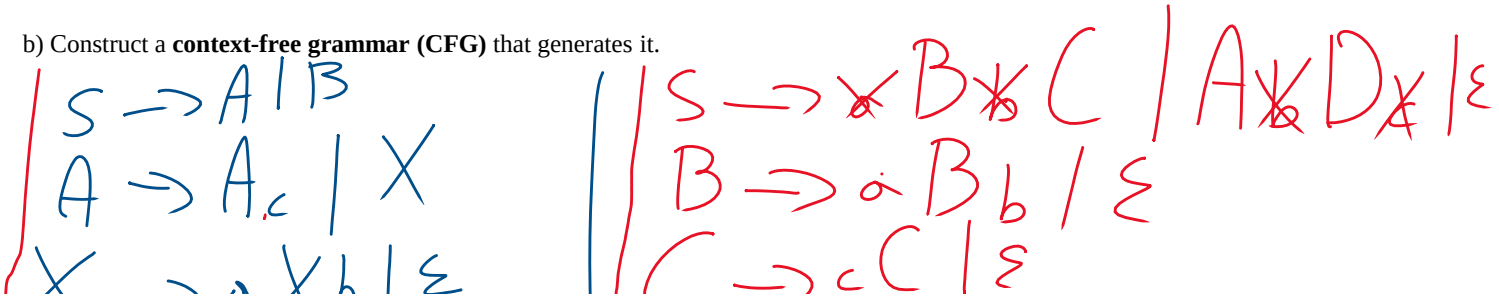
S=XYZ / len XY is <= P so Y is totally in a part

Suppose y=a^k k>=1

S'=Xy^iZ in L and I is >=0

If i=0, S'=a^(p-k) b^p c^q here I is not equal j and j is not equal k

b) Construct a **context-free grammar (CFG)** that generates it.



$X \to aXb \mid \varepsilon$

$B \to aB \mid Y$

$Y \to bYc \mid \varepsilon$

$C \to cC \mid \varepsilon$

$A \to aA \mid \varepsilon$

$D \to bDc \mid \varepsilon$

$\cancel{A \to aAb \mid B \mid \varepsilon}$

$\cancel{B \to cB \mid \varepsilon}$

$A \to BC$

$\cancel{a\,A\,b\,B}$

$aa\varepsilon b\,Bb\,B$

Problem 6 – CFG to Language Description

Given the CFG:

S→aSb | bSa | SS | ab | ba

a) Describe in plain words the language generated.

b) Prove or disprove that this language is **regular**.

$S \to S_1 \mid S_2$

$S_1 \to AB$

$A \to aAb \mid \varepsilon$

$B \to cB \mid \varepsilon$

S_2